

TOYAMA, Japan

WMO: 476070

Lat: 36.7094N

Lon: 137.2030E

Elev: 17

StdP: 101.12

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.4	(c) -1.5	(d) -6.6	(e) 2.2	(f) 2.9	(g) -5.4	(h) 2.4	(i) 2.7	(j) 9.4	(k) 7.4	(l) 8.5	(m) 6.8	(n) 2.9	(o) 200	(p) 0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.3	(c) 34.0	(d) 25.5	(e) 32.7	(f) 25.3	(g) 31.4	(h) 24.9	(i) 26.7	(j) 31.5	(k) 26.1	(l) 30.6	(m) 25.5	(n) 29.7	(o) 4.4	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	29.2	24.9	20.0	28.6	24.3	19.3	28.1	84.1	31.8	81.2	30.7	78.7	29.9	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.7	(c) 6.5	(e) -4.2	(f) 36.6	(g) 1.2	(h) 1.2	(i) -5.1	(j) 37.4	(k) -5.8	(l) 38.2	(m) -6.4	(n) 38.8	(o) -7.3	(p) 39.7		
(a) 9.0	(b) 7.7	(c) 6.5	(e) -4.9	(f) 27.6	(g) 1.1	(h) 0.9	(i) -5.7	(j) 28.2	(k) -6.4	(l) 28.7	(m) -7.0	(n) 29.2	(o) -7.8	(p) 29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	3.0	3.6	7.1	12.4	17.9	21.6	26.0	27.4	23.1	17.3	11.3	5.9
	DBStd	8.87	2.50	2.93	3.48	3.68	3.02	2.45	2.69	2.41	2.94	2.93	3.30	3.13
	HDD10.0	674	216	182	105	16	0	0	0	0	0	0	22	133
	HDD18.3	2114	474	414	349	179	44	3	0	0	1	53	210	386
	CDD10.0	2424	0	2	14	89	245	348	497	540	393	228	63	5
	CDD18.3	822	0	0	0	3	31	101	239	282	144	22	1	0
	CDH23.3	6769	0	0	1	28	194	476	2123	2951	915	80	1	0
	CDH26.7	2385	0	0	0	4	32	99	790	1192	255	12	0	0
Wind	WSAvg	2.9	2.8	2.8	3.2	3.3	3.1	2.6	2.6	2.7	2.7	2.8	2.9	2.9
Precipitation	PrecAvg	2223	245	159	150	128	126	168	220	185	206	161	215	262
	PrecMax	2671	420	282	220	237	309	314	524	503	386	305	409	554
	PrecMin	1527	119	62	69	39	50	59	34	41	82	86	142	153
	PrecStd	279	68	54	42	54	65	64	129	111	78	47	55	93
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	16.6	21.3	26.6	29.7	31.4	35.3	35.8	33.6	28.3	22.7	17.6
		MCWB	7.6	11.4	12.3	15.1	18.5	23.3	25.3	25.4	24.8	20.5	15.4	12.0
	2%	DB	10.2	12.8	18.1	23.2	27.4	29.1	33.7	34.3	31.3	25.4	20.4	14.4
		MCWB	6.6	8.2	10.6	14.6	17.3	21.6	25.3	25.7	24.1	19.2	14.8	10.3
	5%	DB	8.6	10.4	15.5	21.1	25.5	27.6	32.3	33.2	29.5	23.6	18.5	12.4
		MCWB	5.6	6.7	9.7	13.5	16.8	21.2	25.1	25.7	23.5	18.4	13.7	9.2
	10%	DB	7.2	8.5	13.2	18.9	23.7	26.1	30.8	32.0	27.8	22.1	16.8	10.8
		MCWB	4.7	5.6	8.6	12.6	16.3	20.6	24.7	25.4	22.9	17.7	13.2	8.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	12.1	13.7	17.9	20.9	24.7	27.2	27.9	26.3	22.1	17.3	13.4
		MCDB	11.5	16.2	18.4	23.4	25.0	28.8	32.3	32.4	30.4	25.4	19.9	15.8
	2%	WB	7.2	8.9	11.8	15.8	19.4	23.6	26.4	27.1	25.2	20.7	15.9	11.1
		MCDB	9.5	11.9	16.3	20.9	24.1	26.9	31.2	31.8	28.9	23.7	19.0	13.4
	5%	WB	6.0	7.2	10.3	14.6	18.4	22.6	25.8	26.5	24.4	19.6	14.7	9.6
		MCDB	8.0	9.9	14.2	19.3	22.8	25.8	30.4	31.0	27.9	22.3	17.5	11.9
	10%	WB	4.9	5.7	9.1	13.4	17.5	21.7	25.2	25.9	23.6	18.6	13.5	8.3
		MCDB	6.9	8.1	12.6	17.6	21.8	24.7	29.5	30.2	26.8	21.2	16.2	10.5
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.5	8.4	9.3	8.9	7.1	6.9	7.3	7.0	7.7	7.5	6.2
		MCDBR	8.2	10.3	12.7	13.2	12.1	9.7	9.2	9.1	9.1	9.3	10.1	9.2
		MCWBR	5.5	6.9	7.4	6.6	5.3	4.1	3.5	3.7	3.9	4.7	5.9	6.0
	5% WB	MCDBR	7.4	9.6	10.7	10.6	9.7	7.8	8.2	8.1	7.8	7.5	8.4	8.3
MCWBR		5.3	6.9	7.0	6.0	5.0	3.8	3.5	3.7	3.8	4.2	5.5	6.0	
Clear-Sky Solar Irradiance	taub	0.411	0.440	0.504	0.531	0.534	0.557	0.545	0.531	0.478	0.441	0.418	0.410	
	taud	1.963	1.905	1.835	1.870	1.929	1.951	2.048	2.091	2.199	2.197	2.158	2.038	
	Ebn at Noon	732	766	756	764	769	750	756	756	777	766	731	704	
	Edh at Noon	138	165	195	198	190	186	167	157	134	123	114	121	
All-Sky Solar Radiation	RadAvg	1.65	2.47	3.53	4.57	5.25	4.96	4.73	4.89	3.85	3.10	2.07	1.43	
	RadStd	0.17	0.20	0.28	0.38	0.57	0.37	0.72	0.52	0.29	0.27	0.18	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.65	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page